

LANGUAGE: JAVA PROGRAMMING LEVEL:INTERMEDIATE-ADVANCED

Day	Topic	Subtopics	Interview Coding Problems (LeetCode / HackerRank)	Intermediate–Advanced Corporate Use Case
Day 1	Advanced Java Fundamentals	JVM Internals, Memory Management, Garbage Collection Basics, Data Types, Wrapper Classes, String Interning, Operators, Control Statements	LeetCode: Palindrome Number, Reverse Integer, Valid Anagram HackerRank: Java Datatypes, Java Strings Introduction	Enterprise HR & Payroll Processing Engine
Day 2	Advanced Arrays, Strings & Problem Solving	Arrays, 2D Arrays, Sliding Window, Prefix Sum, StringBuilder, Mutable vs Immutable Objects, Command Line Arguments	LeetCode: Longest Substring Without Repeating Characters, Maximum Subarray, Product of Array Except Self HackerRank: Sparse Arrays, Left Rotation	Real-Time Customer Feedback Analytics System
Day 3	OOP Design Principles	Classes & Objects, Constructors, Encapsulation, Inheritance, Polymorphism, Abstraction, Interfaces, SOLID Principles	LeetCode: Design Parking System, Merge Intervals HackerRank: Inheritance, Abstract Classes	Enterprise Banking & Loan Approval System
Day 4	Advanced OOP & Design Patterns	Runtime Polymorphism, Dynamic Binding, Composition vs Inheritance, Singleton, Factory Pattern, Strategy Pattern, Dependency Injection Basics	LeetCode: LRU Cache, Climbing Stairs HackerRank: Java Factory Pattern	Insurance Claim Processing Platform
Day 5	Exception Handling & Logging	Exception Hierarchy, Custom Exceptions, try-with-resources, Assertions, Logging Framework Basics, Debugging Techniques	LeetCode: Contains Duplicate, Missing Number HackerRank: Java Exception Handling	Secure Financial Transaction Validation System
Day 6	Collections Framework & Generics	ArrayList, LinkedList, Stack, Queue, HashMap, TreeMap, HashSet, Generics, Comparable vs Comparator	LeetCode: Group Anagrams, Top K Frequent Elements, Two Sum HackerRank: Java Map, Java Priority Queue	Fraud Detection & Transaction Monitoring Engine
Day 7	Advanced Collections & Java 8 Features	Stream API, Lambda Expressions, Functional Interfaces, Optional Class, Method References, Parallel Streams	LeetCode: K Closest Points to Origin, Merge Sorted Array HackerRank: Java Lambda Expressions	Real-Time Recommendation System
Day 8	Multithreading & Concurrency	Threads, Runnable, Callable, Executor Framework, Synchronization, Deadlocks, Race	LeetCode: Print in Order, FizzBuzz Multithreaded, Building H2O HackerRank: Java	Multi-threaded High Volume Log Processing System

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		Conditions, Thread Pooling	Threads	
Day 9	File Handling, Serialization & Networking	File Streams, Buffered Streams, Serialization, Deserialization, Socket Programming Basics, TCP/IP Concepts	LeetCode: Number of Islands, Flood Fill HackerRank: Java Networking Basics	Distributed File Synchronization System
Day 10	JDBC & Database Integration	JDBC Architecture, CRUD Operations, PreparedStatement, Batch Processing, Transactions, Connection Pooling Basics	LeetCode: Rotate Array, Binary Search HackerRank: SQL Joins, SQL Aggregations	Enterprise Resource Planning (ERP) Database System
Day 11	Advanced Java & System Design Concepts	Immutable Classes, Reflection API, Annotations, Memory Optimization, JVM Tuning Basics, Microservices Introduction	LeetCode: Valid Sudoku, Search in Rotated Sorted Array HackerRank: Java Reflection	Scalable Microservices-Based E-Commerce Backend
Day 12	Advanced Interview Coding & DSA	Time Complexity, Space Complexity, Sliding Window, Hashing, Recursion, Backtracking, Searching & Sorting Optimization	LeetCode: Longest Substring Without Repeating Characters, Container With Most Water, Merge Intervals, Binary Search, Top K Frequent Elements HackerRank: Balanced Brackets, Sherlock and Arrays, Problem Solving (Medium)	Enterprise Coding Assessment & Evaluation Platform
Day 13	Advanced Coding + RDBMS Concepts	First Half – Advanced Coding: Mock Coding Interviews, Optimization Techniques, Code Review, Debugging & Refactoring. Second Half – RDBMS: Normalization, Indexing, Transactions, ACID Properties, Joins, Stored Procedures, Views, SQL Optimization	LeetCode: LRU Cache, Best Time to Buy & Sell Stock, Valid Parentheses HackerRank: SQL Advanced Select, Weather Observation Station, Dynamic Array	Enterprise Data Warehouse & Reporting System
Day 14	Cloud Basics, GitHub, Docker, Kubernetes & CI/CD	Cloud Computing, IaaS/PaaS/SaaS, GitHub Workflow, Branching Strategy, Docker Images & Containers, Docker Compose, Kubernetes Pods & Deployments, CI/CD Pipelines, GitHub Actions, Jenkins Basics,	LeetCode: Single Number, Move Zeroes, Merge Sorted Array HackerRank: Linux Shell Basics, Git Tutorials, 30 Days of Code	End-to-End Cloud Native Microservices Deployment Pipeline using Docker + Kubernetes + CI/CD

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		Infrastructure as Code (IaC)		



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